

Hermes Agent Architecture

Public-safe architecture snapshot generated at 2026-07-18T06:15:22-04:00.

Source of truth: local Hermes configuration and runtime status on the operator Linux host.

Sensitive values, private chat IDs, OAuth headers, personal finance/client workflow names, and tokens are intentionally omitted or grouped.

Executive Summary

Hermes Agent runs as a **Linux-primary, multi-platform AI operations hub**. The gateway is a systemd service that connects Telegram, Mattermost, WhatsApp, API Server, Home Assistant, MCP servers, cron jobs, skills, memory/context stores, and remote Windows operations.

The current architecture adds a **durable Kanban orchestration layer**: `default` remains the interactive/orchestrator profile, while `researcher`, `worker`, and `reviewer` are launched as ephemeral Kanban worker-runs for durable research, implementation, and verification tasks. Safe post-session automation generates local review artifacts (session summaries, skill-mining candidates, trace-derived code graphs) without silently installing skills or uploading traces.

The default model remains `openai-codex / gpt-5.5`. Local/experimental providers such as **LM Studio** and **Free Kimi** are configured as optional providers only; they are not the default route for Telegram or production workflows.

High-Level Architecture

[!Hermes Architecture](#)

Runtime Topology

Layer	Components	Notes
User channels	Telegram, Mattermost, WhatsApp, API Server	Gateway adapters route inbound messages into the same Hermes agent core.
Agent core	Hermes Agent gateway + CLI + tool loop	Runs on Linux under <code>hermes-gateway.service</code> .
Models	OpenAI Codex primary, Copilot fallback, optional LM Studio, optional Free Kimi proxy	Provider selection is explicit; experimental endpoints are not default.
Tools	Terminal, files, browser, web, vision, TTS, cron, delegation, Home Assistant, GBrain, NotebookLM, CodeGraph	Enabled toolsets differ by platform/session but the gateway exposes the main operational surface.
Knowledge	GBrain, NotebookLM, session search, imported Claude context, skills	Long-term memory and research context are split across structured stores and markdown skills.
Automation	Hermes cron scheduler, no-agent scripts, hooks, plugins, post-session automation	Backups, monitoring, GitHub publication, finance snapshots, smart-home logging, news/media digests, watchdogs, session summaries, skill-mining reviews, and trace graphs.
Durable orchestration	Kanban dispatcher + <code>default/researcher/worker/reviewer</code> profiles	The gateway dispatcher claims ready tasks and launches profile-specific worker-runs; <code>stopped</code> profiles are normal idle state.
Operational surfaces	Tasks, skills, rules, hooks, plugins, MCP servers, toolsets	Documented as public-safe counts/categories plus selected non-sensitive examples.
Remote systems	Windows workstation over SSH, Home Assistant, Google Drive/GitHub, local LM Studio	Linux remains the control plane; Windows is operated remotely when needed.

Operational Surface Inventory

Surface	Detected public-safe state	Notes
Scheduled tasks / cron	65 jobs; 34 no-agent script jobs; 0 agent-backed jobs	Exact private task names are grouped by category.
Skills	284 detected skill files across 25 categories	Private/client-sensitive skill names are omitted from examples.
Hooks / webhooks	shell allowlist present: False; allowlist entries: 0; plugin hook manifests: 30	Hook command bodies are not published.
Plugins	68 visible plugin rows captured; enabled estimate 6	Descriptions omitted to avoid leaking credential/env surfaces.
MCP servers	10 configured MCP servers	GBrain, NotebookLM, CodeGraph are the active core MCP surfaces.

Scheduled tasks / cron categories

Category	Active jobs	Public-safe purpose
Backup & sync	5	Protect configuration, repositories, databases, and knowledge stores.
GitHub & publishing	6	Maintain GitHub/publication surfaces and repo health digests.
Home automation	2	Log smart-home/home-environment telemetry.
Knowledge & memory	6	Keep GBrain/memory/context stores healthy and up to date.
Media/news monitoring	3	News, RSS, YouTube, and briefing pipelines.
Other scheduled automation	28	Other local automation jobs.
Private finance automation	5	Private finance workflow snapshots; details omitted from public docs.
Reliability watchdogs	10	Auto-healing, environment guards, timeout/watchdog checks.

Skills surface

Hermes currently has a broad skill surface. The public inventory lists category counts and non-sensitive examples only.

Skill category	Count
.archive	11
android	2
apple	5
autonomous-ai-agents	11
creative	37
data-science	2
devops	10
ecc-imports	4
email	3
gaming	2
github	14
mcp	2
media	7
mlops	24
note-taking	4
personal	8
productivity	27
red-teaming	1
research	20
security	3
smart-home	5
social-media	2
software-development	58
uncategorized	21
web-development	1

Public-safe skill examples:

Skill	Category	Public-safe description
dspy	mlops	DSPy: declarative LM programs, auto-optimize prompts, RAG.
fine-tuning-with-trl	mlops	TRL: SFT, DPO, PPO, GRPO, reward modeling for LLM RLHF.
axolotl	mlops	Axolotl: YAML LLM fine-tuning (LoRA, DPO, GRPO).
unsloth	mlops	Unsloth: 2-5x faster LoRA/QLoRA fine-tuning, less VRAM.
instructor	mlops	Extract structured data from LLM responses with Pydantic validation, retry failed extractions automatically, parse complex JSON with type sa
nemo-curator	mlops	GPU-accelerated data curation for LLM training. Supports text/image/video/audio. Features fuzzy deduplication (16× faster), quality filterin
huggingface-hub	mlops	HuggingFace hf CLI: search/download/upload models, datasets.
segment-anything-model	mlops	SAM: zero-shot image segmentation via points, boxes, masks.
chroma	mlops	Open-source embedding database for AI applications. Store embeddings and metadata, perform vector and full-text search, filter by metadata.
faiss	mlops	Facebook
weights-and-biases	mlops	W&B: log ML experiments, sweeps, model registry, dashboards.
evaluating-llms-harness	mlops	lm-eval-harness: benchmark LLMs (MMLU, GSM8K, etc.).
llama-cpp	mlops	llama.cpp local GGUF inference + HF Hub model discovery.
obliteratus	mlops	OBLITERATUS: obliterate LLM refusals (diff-in-means).
serving-llms-vllm	mlops	vLLM: high-throughput LLM serving, OpenAI API, quantization.
outlines	mlops	Outlines: structured JSON/regex/Pydantic LLM generation.
qdrant-vector-search	mlops	High-performance vector similarity search engine for RAG and semantic search. Use when building production RAG systems requiring fast neares
strategic-reading	uncategorized	Read a book, article, transcript, or case study through the lens of a specific strategic problem you
gif-search	media	Search/download GIFs from Tenor via curl + jq.
spotify	media	Spotify: play, search, queue, manage playlists and devices.

Skill	Category	Public-safe description
youtube-content	media	YouTube transcripts to summaries, threads, blogs.
article-enrichment	uncategorized	Transform raw article text dumps in the brain into structured pages with executive summary, verbatim quotes, key insights, why-it-matters, a
osint-investigation	research	Public-records OSINT investigation framework — SEC EDGAR filings, USAspending contracts, Senate lobbying, OFAC sanctions, ICIJ offshore leak
domain-intel	research	Passive domain reconnaissance using Python stdlib. Subdomain discovery, SSL certificate inspection, WHOIS lookups, DNS records, domain avail
[REDACTED]	research	Evaluate agent memory systems locally before adopting cloud memory vendors like Synap, Mem0, Zep, or Supermemory.

Hooks, webhooks, and plugin hook manifests

Hermes has multiple hook-related surfaces: shell-hook allowlists, webhook subscriptions, and imported Claude/Claude plugin hook manifests. The public repo records only surface/count information, not hook command bodies.

Hook manifest surface
<code>.hermes/oss-evals/2026-06-26/orca-sandbox/home/.gemini/config/hooks.json</code>
<code>.hermes/oss-evals/2026-06-26/orca-sandbox/home/.cursor/hooks.json</code>
<code>.hermes/oss-evals/2026-06-26/orca-sandbox/config/orca/codex-runtime-home/home/hooks.json</code>
<code>.hermes/oss-evals/agent-orchestrators/2026-07-13/wit/hooks/hooks.json</code>
<code>.hermes/hermes-agent.pre-v2026.7.7-20260707_215257_EDT/.codex/hooks.json</code>
<code>.claude/plugins/marketplaces/thedotmack/plugin/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/thedotmack/cursor-hooks/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-plugins-official/plugins/learning-output-style/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-plugins-official/plugins/security-guidance/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-plugins-official/plugins/hookify/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-plugins-official/plugins/ralph-loop/hooks/hooks.json</code>
[REDACTED]
<code>.claude/plugins/marketplaces/claude-code-warp/plugins/warp/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/headroom-marketplace/plugins/headroom-agent-hooks/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-code-plugins/plugins/learning-output-style/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-code-plugins/plugins/security-guidance/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-code-plugins/plugins/hookify/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-code-plugins/plugins/ralph-wiggum/hooks/hooks.json</code>
[REDACTED]
<code>.claude/plugins/marketplaces/ponytail/hooks/hooks.json</code>

Plugin surface

Plugin	Status
browser-firecrawl	not enabled
chronos	not enabled
basic	not enabled
drain	not enabled
nous	not enabled
self-hosted	not enabled
disk-cleanup	not enabled
google_meet	not enabled
fal	not enabled
krea	not enabled
openai	not enabled
openai-codex	not enabled
openrouter	not enabled
xai	not enabled
alibaba-provider	not enabled
anthropic-provider	not enabled
arcee-provider	not enabled
bedrock-provider	not enabled
copilot-provider	not enabled
custom-provider	not enabled
deepseek-provider	not enabled
gemini-provider	not enabled
gmi-provider	not enabled
kilocode-provider	not enabled
minimax-provider	not enabled

Plugin	Status
nous-provider	not enabled
novita-provider	not enabled
nvidia-provider	not enabled
stepfun-provider	not enabled
vertex-provider	not enabled

Low-Level Surface Files

The repository includes dedicated, low-level public-safe files for each operational surface:

File	Contents
docs/surfaces/tasks.md	Scheduled tasks/cron jobs, modes, scripts, schedules, delivery class, workdir class.
docs/surfaces/skills.md	Skill categories, counts, public-safe examples, operational semantics.
docs/surfaces/hooks.md	Shell hook allowlist state, webhook state, plugin hook manifests.
docs/surfaces/plugins.md	Hermes plugin registry rows and status.
docs/surfaces/mcp-and-toolsets.md	MCP servers and toolset count estimates.
docs/surfaces/models-and-gateway.md	Model routing, gateway status, channel/platform surface.
docs/surfaces/orchestration.md	Kanban profiles, rules layer, self-improvement automation, and operating contract.

Model Routing

Role	Provider	Model	Notes
Primary	openai-codex	gpt-5.6-sol	Default for Telegram/API/CLI gateway sessions
Fallback			Used when primary fails
Optional provider	lmstudio	qwenvl3bunc	http://127.0.0.1:1234/v1
Optional provider	nvidia	meta/llama-3.3-70b-instruct	https://integrate.api.nvidia.com/v1
Optional provider	freekimi	cfbt-kimi	http://127.0.0.1:3271/v1
Optional provider	forge_freekimi	cfbt-kimi	http://127.0.0.1:8081/v1
Optional provider	forge_lmstudio	qwenvl3bunc	http://127.0.0.1:8082/v1
Optional provider	chatgpt_web	chatgpt-5.6-sol-high-web	https://codex.guber.dev/v1

Local model trial status

Item	Status
LM Studio endpoint	available at http://127.0.0.1:1234/v1
Reported model IDs	qwenvl3bunc, mythosnanoq6, qwythos9bq5, gemma4coderq3, gemma4coderq4, oymuncq4, [REDACTED], gemma4unc, [REDACTED]
Chat smoke test	`blocked_or_unavailable: {

"error": { "message": "Failed to load model \"gemma4unc\". Error: Model loading was stopped due to insufficient system resources. Under the current settings, this model requires approximately 14.36 GB of memory, and continuing | | Safety decision | Main Hermes remains openai-codex/gpt-5.5`; local provider is optional until a model can load reliably. |

MCP and External Tooling

MCP server	Transport	Public-safe purpose
gbrain	Local HTTP MCP on 127.0.0.1:3131/mcp	Personal knowledge graph, memory, code graph, facts, schema tools.
notebooklm	npx notebooklm-mcp@latest	Grounded research over registered Google NotebookLM notebooks.
codegraph	Local Node command	Code intelligence over selected repositories.

Scheduled Automation

Category	Active jobs	Public-safe purpose
Backup & sync	5	Protect configuration, repositories, databases, and knowledge stores.
GitHub & publishing	6	Maintain GitHub/publication surfaces and repo health digests.
Home automation	2	Log smart-home/home-environment telemetry.
Knowledge & memory	6	Keep GBrain/memory/context stores healthy and up to date.
Media/news monitoring	3	News, RSS, YouTube, and briefing pipelines.
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Agentic Operating Model

Item	Value	Meaning
Interactive orchestrator	default	Main Telegram/API/CLI profile; decides whether to answer directly, delegate, create Kanban work, or schedule cron.
Research lane	researcher	Evidence-backed OSS/docs/web due diligence and adoption recommendations.
Execution lane	worker	File/terminal/code/config implementation and verification for durable work items.
Review lane	reviewer	Independent review, regression checks, security/config sanity, and final validation.
Default assignee	worker	Fallback when a task/decomposer does not choose a specialist.
Failure limit	5	Raised above the previous aggressive default to avoid noisy retry failures.
Per-profile concurrency	1	Keeps one profile from saturating model/API/browser capacity.
Post-session automation	enabled	Generates review artifacts only; does not auto-install remote skills or mutate active prompts.

The important runtime distinction is that `researcher`, `worker`, and `reviewer` do **not** need to be continuously running gateway profiles. They are idle until a Kanban task is assigned to them; then the gateway dispatcher starts a worker-run for that profile and the worker completes, blocks, or retries the task.

Current Profiles

The live system currently exposes the public-safe profile roster as:

Profile	Model	Gateway	Alias	Distribution
◆default	gpt-5.6-sol	running	—	—
claude	gpt-5.6-sol	stopped	hermes-claude	—
coding	gpt-5.6-sol	stopped	coding	—
researcher	gpt-5.6-sol	stopped	hermes-researcher	—
reviewer	gpt-5.6-sol	stopped	hermes-reviewer	—
security-restricted	gpt-5.6-sol	stopped	hermes-security	—
worker	gpt-5.6-sol	stopped	hermes-worker	—

Current profile contract:

Profile	Role	Runtime behavior
default	Interactive orchestrator	Stays running in Telegram/API/CLI gateway; routes work.
researcher	Research/due-diligence worker	Normally stopped; launched by Kanban for evidence gathering.
worker	Implementation/execution worker	Normally stopped; fallback assignee for durable tasks.
reviewer	Independent verification worker	Normally stopped; launched for review/regression/security checks.

Reliability and Safety Boundaries

- Public repo does **not** publish secrets, chat IDs, API keys, OAuth tokens, cookies, private finance details, or client-specific payloads.
- The gateway is supervised by systemd and multiple watchdog cron jobs.
- GBrain runs through a shared local HTTP MCP server to avoid PGLite lock contention.
- Backups and sync jobs run through cron with Google Drive/GitHub targets.
- Windows is treated as a remote worker controlled via SSH, not as the primary Hermes control plane.
- Experimental providers are explicitly optional and must pass direct API + Hermes smoke tests before being used for real agent work.

Operational Notes

- Hermes version/status summary:

```
Hermes Agent v0.18.2 (2026.7.7.2) · upstream 5402cb55 · local db8ab0e3 (+12 carried commits)
Install directory: ~/.hermes/hermes-agent
Install method: git
Python: 3.11.15
OpenAI SDK: 2.24.0
Update available: 1142 commits behind – run 'hermes update'
```

- Fallback chain:

```
[REDACTED]
```

- MCP list:

MCP Servers:

Name	Transport	Tools	Status
codegraph	~/nvm/versions/no... all	enabled	
gbrain	http://127.0.0.1:3131/mcp	all	enabled
notebooklm	npm -y notebooklm-mcp@latest	all	enabled
windows-cua	~/local/bin/windo... all	enabled	
monarch	https://api.monarch.com/mcp	17 selected	enabled
cloudflare-api	https://mcp.cloudflare.co... all	enabled	
pixelrag	~/github/pixelrag-... all	enabled	
vibe_trading	~/hermes/scripts/... 29 selected	enabled	
display	https://api.display.dev/v... all	enabled	
open_notebook	~/hermes/oss-eval... all	enabled	

Maintenance

This repository is regenerated by `scripts/update_architecture.py`. The updater is designed to be safe for a public repository:

- 1. collect local Hermes runtime state,
- 2. redact secrets and private identifiers,
- 3. group sensitive scheduled jobs by category,
- 4. regenerate Markdown, SVG, HTML, JSON inventory, and PDF,
- 5. commit and push only if files changed.

Source Files

- `ARCHITECTURE.md` - canonical public architecture document.
- `docs/hermes-architecture.svg` - standalone diagram.
- `docs/hermes-architecture.html` - browser-friendly diagram page.
- `docs/Hermes-Architecture.pdf` - rendered PDF copy.
- `docs/surfaces/*.md` - low-level public-safe inventories for tasks, skills, hooks, plugins, MCP/toolsets, and gateway/model routing.
- `data/inventory.public.json` - redacted machine-readable snapshot.
- `scripts/update_architecture.py` - public-safe regeneration script.

Rendered from GBrain markdown via Hermes brain-pdf.